

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only):**[2 x 7 = 14]**

- (a) For a base current of $10\mu\text{A}$, what is the value of collector current in common emitter if $\beta_{ac} = 100$?
 - (i) $10\mu\text{A}$
 - (ii) $100\mu\text{A}$
 - (iii) 1mA
 - (iv) 10mA
- (b) If an amplifier with gain of -1000 and feedback factor $\beta = -0.1$ had a gain change of 20% due to temperature, the change in gain of the feedback amplifier would be
 - (i) 10%
 - (ii) 5%
 - (iii) 0.2%
 - (iv) 0.01%
- (c) A trivalent impurity has Valence electrons.
 - (i) 4
 - (ii) 5
 - (iii) 6
 - (iv) 3
- (d) Zener diodes are used primarily as
 - (i) Amplifiers
 - (ii) Voltage regulators
 - (iii) Rectifiers
 - (iv) Oscillators
- (e) Peak inverse voltage of diode used in Half-wave rectifier is
 - (i) $2V_m$
 - (ii) $V_m/2$
 - (iii) V_m
 - (iv) $V_m/3$
- (f) For every 10°C increase in temperature, the reverse saturation current of a p-n junction will be increased by:
 - (i) 10 times
 - (ii) 2 times
 - (iii) 4 times
 - (iv) Remain same
- (g) A BJT transistor operates in which region when the base-emitter junction is forward-biased and the base-collector junction is reverse-biased?
 - (i) Active region
 - (ii) Saturation region
 - (iii) Cutoff region
 - (iv) Reverse active region
- (h) In an RC phase shift oscillator, the phase shift provided by each RC stage is :
 - (i) 30°
 - (ii) 45°
 - (iii) 60°
 - (iv) 90°
- (i) If the PIV rating of a diode is exceeded
 - (i) the diode conducts poorly
 - (ii) the diode is destroyed
 - (iii) the diode behaves as Zener diode
 - (iv) None of the above
- (j) For $I_{DSS} = 9\text{mA}$ and $V_p = -3.5\text{V}$, I_D for $V_{GS} = 0\text{V}$ is
 - (i) 8mA
 - (ii) 9mA
 - (iii) 10mA
 - (iv) 11mA

Q.2 (a) Define the following:

- (i) Common mode rejection ratio (CMRR)
- (ii) Gain bandwidth product
- (iii) Slew rate of op-amp

[7]

- (b) State the Barkhausen condition for an electronic system to oscillate with feedback. [7]
- Q.3 (a) Derive the expression for stability factor for fix bias circuit with respect to I_{CO} , V_{BE} and β . [7]
- (b) A voltage divider biased circuit has $R_1=39k\Omega$, $R_2=82k\Omega$, $R_C=3.3k\Omega$, $R_E=1k\Omega$ and $V_{CC}=18V$. The silicon transistor has used $\beta = 120$. Find Q - point and stability factor. [7]
- Q.4 (a) With the help of block diagram, explain the concept of feedback. [7]
- (b) Discuss with the help of circuit example, the purpose of providing – [7]
- (i) negative feedback;
- (ii) positive feedback in amplifier.
- Q.5 (a) With a neat circuit diagram and waveforms, explain the working of full wave bridge rectifier and show that its ripple factor is 0.48. [7]
- (b) Describe in detail the avalanche and Zener breakdown mechanism in Zener diode. [7]
- Q.6 (a) Derive the expression for output voltage of an instrumentation amplifier. Also write its advantages and disadvantages [8]
- (b) For an N-channel MOSFET the parameters given as $\mu C_{ox} \frac{W}{L} = \frac{0.2mA}{V^2}$, $V_{DS} = 0.2V$, and $V_t = 0.7V$. Find the region of operation and the drain current. [6]
- Q.7 (a) Differentiate between JFET and BJT indicating the advantages and disadvantages. [7]
- (b) What is faithful amplification? Explain the conditions to be fulfilled to achieve faithful amplification in transistor amplifier. [7]
- Q.8 (a) Draw and explain the pin configuration of a 741 Op-Amp. Also explain the internal structure of an Op-Amp with the help of block diagram. [7]
- (b) For the circuit shown below. Assume zener voltage to be 4.78v and voltage drop across the forward biased zener to be 0.7v. Find the peak voltage of output. [7]
- Q.9 (a) State the characteristics of an ideal transformer. [2]
- (b) Define *rms* value, form factor, peak factor, complex power and half power frequency. [5]
- (c) Two two-port network a and b, with open-circuit impedances Z_a and Z_b are connected in series. Drive the Z-parameter equations. [7]

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Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2023

Course: B.Tech

Code: 100302

Subject: Analog Electronic Circuits

Time: 03 Hours

Full Marks: 70

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Q.1 Answer the following questions (any seven only):

[2 x 7 = 14]

- (a) Draw the pn junction diode VI characteristics.
- (b) Mention the advantages of Full Wave Rectifier.
- (c) What are the different configurations of BJT?
- (d) What is stability factor?
- (e) What is MOSFET? Classify the types of MOSFET.
- (f) What is meant by pinchoff voltage?
- (g) Define an operational amplifier.
- (h) Define Slew rate.
- (i) Mention some of the non-linear applications of op – amp.
- (j) List the features of instrumentation amplifier.

Q.2 (a) What is a PN Junction? Explain the formation of depletion layer in a PN junction.

[7]

- (b) Draw the circuit diagram of half wave rectifier and explain its operation with the help of waveforms.

[7]

Q.3 (a) If the base current in a transistor is $20\mu\text{A}$ when the emitter current is 6.4mA , what are the values of α and β ? Also calculate the collector current.

[7]

- (b) Why hybrid model is used for the analysis of BJT amplifier at low frequencies? Draw the hybrid model for CE transistor and derive the parameters.

[7]

Q.4 (a) Draw and explain the small signal model of FET at low frequency.

[7]

- (b) With the help of neat diagram, explain the operation and characteristics of n-channel enhancement type MOSFET.

[7]

Q.5 (a) What are the factors that affect the frequency stability of an oscillator? How frequency stability can be improved in oscillators.

[7]

- (b) Compare different configurations of differential amplifier.

[7]

Q.6 (a) Write the design steps of the first order low pass filter and draw its circuit.

[7]

- (b) With the help of neat and clear sketch, explain the working of a MOS transistor as a switch.

[7]

Q.7 Explain the basic building block diagram of an op-amp. Write short note on input bias current, output bias current, slew rate and output offset voltage of an op-amp.

[14]

Q.8 (a) Explain the working of series clipping circuit to clip the input sinusoidal signal above reference level. Draw the waveforms and transfer characteristics.

[7]

- (b) Compare BJT and FET on different parameters.

[7]

Q.9 (a) Write Short notes on (i) Zero Crossing Detector (ii) Precision Rectifier

[7]

- (b) Explain the function of Square-wave and triangular –wave generators.

[7]

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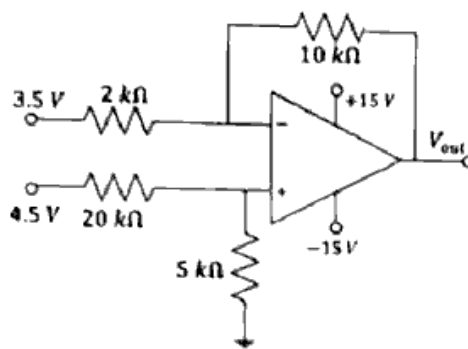
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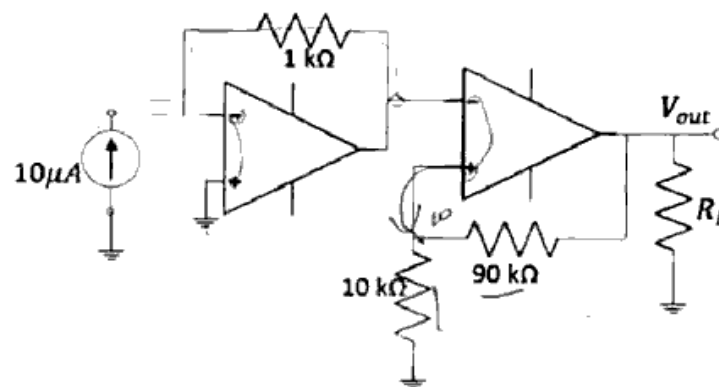
[2 x 7 = 14]

- (a) A Zener diode rated 5.1V is used with a 220 Ω resistor. If input is 12V, what is the Zener current?
 - (i) 31.4 mA
 - (ii) 50 mA
 - (iii) 20 mA
 - (iv) 10 mA
- (b) At room temperature, a transistor has a collector current of 5 mA and $V_{BE} = 0.7V$. Find g_m .
 - (i) 7.14 S
 - (ii) 194 mS
 - (iii) 5 S
 - (iv) 3.5 S
- (c) Which region does a MOSFET operate in for amplification?
 - (i) Cutoff
 - (ii) Triode
 - (iii) Saturation
 - (iv) Ohmic
- (d) The trans-conductance (g_m) of a MOSFET is a measure of:
 - (i) Voltage gain
 - (ii) Input impedance
 - (iii) Output conductance
 - (iv) Change in drain current per unit change in gate voltage
- (e) Which parameter limits the maximum frequency response of an op-amp?
 - (i) Slew rate
 - (ii) Offset voltage
 - (iii) Input bias current
 - (iv) Common-mode gain
- (f) An inverting amplifier has $R_f = 100\text{ k}\Omega$ and $R_{in} = 10\text{ k}\Omega$. What is the voltage gain?
 - (i) -100
 - (ii) -10
 - (iii) 10
 - (iv) 1
- (g) Which op-amp circuit converts a sinusoidal input to a square wave output?
 - (i) Peak detector
 - (ii) Comparator
 - (iii) Schmitt trigger
 - (iv) Integrator
- (h) A MOSFET has $I_D = 2\text{ mA}$ and $V_{GS} = 2.5$. If $V_{th} = 1\text{ V}$ and $k_n = 1\text{ mA/V}^2$, what is the value of I_D .
 - (i) 1 mA
 - (ii) 2 mA
 - (iii) 3.5 mA
 - (iv) 4 mA
- (i) Which component is used to maintain a constant voltage across the load?
 - (i) P-N junction diode
 - (ii) Zener diode
 - (iii) Schottky diode
 - (iv) Tunnel diode
- (j) Which configuration of BJT offers the highest current gain?
 - (i) Common Base
 - (ii) Common Collector
 - (iii) Common Emitter
 - (iv) Emitter Follower

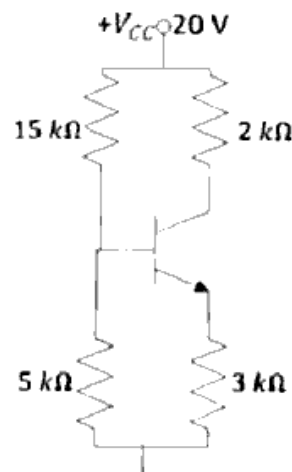
- Q.2 (a) Explain the working of a differential amplifier using BJTs. Derive expressions for differential and common-mode gain. [7]
- (b) Assume the op-amp is ideal, find the output voltage V_{out} . [7]



- Q.3 (a) Design and analyze a Wein Bridge Oscillator using op-amp. Derive the expression for frequency of oscillation. [7]
- (b) Explain the operation of a Zener diode as a voltage regulator. Design a circuit to provide 5V regulated output. [7]
- Q.4 (a) Explain the operation of a hysteretic comparator using op-amp. Draw and analyze the transfer characteristics. [7]
- (b) Assume the op-amp is ideal, find the output voltage V_{out} . [7]



- Q.5 (a) Design and analyze a clamping circuit. Explain how it shifts the DC level of an input waveform. [7]
- (b) Explain the effect of channel length modulation on the performance of a MOSFET amplifier. [7]
- Q.6 (a) Design a triangular wave generator using op-amps. [7]
- (b) For the circuit shown below, draw D.C. load line. Assume β is very large. [7]



- Q.7 (a) Design a first-order low-pass butter worth filter circuit using op-amp and derive the expression of high cut off frequency and transfer function of the filter. [7]
(b) Explain the role of op-amp in waveform shaping circuits with examples. [7]
- Q.8 (a) Derive the expression for ripple voltage and peak inverse voltage in a full-wave rectifier. [7]
(b) Explain the various biasing techniques for a BJT amplifier. Suggest which one is better and why? [7]
- Q.9 Write short notes on *any two* of the following:- [7 x 2 = 14]
(a) Current mirror using BJT
(b) Trans-conductance of MOSFET
(c) Junction break-down in Zener diodes
(d) high-frequency equivalent circuit of BJT
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